Casimir's Trick for $\mathcal{M}_s\mathcal{M}_t^*$: Every Step Shown Explicitly

PHYS 7502

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1 The Two Amplitudes (Professor's Convention)

Convention: $1 = e^-_{\text{in}}$, $2 = \gamma_{\text{in}}$, $3 = \gamma_{\text{out}}$, $4 = e^-_{\text{out}}$. Setting $m = 0$:

$$\mathcal{M}_s = \frac{e^2}{s}\, \varepsilon^*_{3\alpha}\, \varepsilon_{2\beta} \Big[\bar{u}(4)\, \gamma^\alpha\, (\not{p}_1 + \not{p}_2)\, \gamma^\beta\, u(1)\Big] \tag{1}$$

$$\mathcal{M}_t^* = \frac{e^2}{t}\, \varepsilon^*_{2\mu}\, \varepsilon_{3\nu} \Big[\bar{u}(1)\, \gamma^\nu\, (\not{p}_1 - \not{p}_3)\, \gamma^\mu\, u(4)\Big] \tag{2}$$

(In $\mathcal{M}_t^*$ the spinors have swapped, the γ-matrices reversed order, and the polarizations conjugated relative to $\mathcal{M}_t$.)

2 Forming the Product $\mathcal{M}_s\mathcal{M}_t^*$

Multiplying (1) and (2):

$$\mathcal{M}_s\mathcal{M}_t^* = \frac{e^4}{st}\, \varepsilon^*_{3\alpha}\varepsilon_{3\nu}\, \varepsilon_{2\beta}\varepsilon^*_{2\mu} \times \underbrace{\Big[\bar{u}(4)\, \gamma^\alpha(\not{p}_1 + \not{p}_2)\gamma^\beta\, u(1)\Big]}_{A}\, \underbrace{\Big[\bar{u}(1)\, \gamma^\nu(\not{p}_1 - \not{p}_3)\gamma^\mu\, u(4)\Big]}_{B} \tag{3}$$

We need to sum over spins of particles 1 and 4, then sum over photon polarizations of particles 2 and 3. Let us do the **spin sum first** using Casimir's trick, then the **photon sum** afterward.

3 Casimir's Trick: The Key Idea

Casimir's Trick (Griffiths Eq. 7.125)

For any two 4×4 matrices Γ_1 and Γ_2:

$$\sum_{\text{all spins}} \Big[\bar{u}(a)\, \Gamma_1\, u(b)\Big]\Big[\bar{u}(b)\, \Gamma_2\, u(a)\Big] = \text{Tr}\Big[\Gamma_1\, (\not{p}_b + m_b)\, \Gamma_2\, (\not{p}_a + m_a)\Big] \tag{4}$$

The trick works because:

1. The first bilinear has $\bar{u}(a) \cdots u(b)$.
2. The second bilinear has $\bar{u}(b) \cdots u(a)$.
3. This "*a*-*b*-*b*-*a*" pattern lets you use completeness $\sum u(b)\bar{u}(b) = \not{p}_b + m_b$ to stitch the two bilinears together into a single trace.

Identifying a, b, Γ_1, Γ_2 in Our Problem

Looking at the product $A \times B$:

$$A = \bar{u}(\underbrace{4}_{a})\ \underbrace{\gamma^\alpha(\not{p}_1 + \not{p}_2)\gamma^\beta}_{\Gamma_1}\ u(\underbrace{1}_{b}) \tag{5}$$

$$B = \bar{u}(\underbrace{1}_{b})\ \underbrace{\gamma^\nu(\not{p}_1 - \not{p}_3)\gamma^\mu}_{\Gamma_2}\ u(\underbrace{4}_{a}) \tag{6}$$

So: $a =$ particle 4, $b =$ particle 1, and:

$$\Gamma_1 = \gamma^\alpha(\not{p}_1 + \not{p}_2)\gamma^\beta \tag{7}$$

$$\Gamma_2 = \gamma^\nu(\not{p}_1 - \not{p}_3)\gamma^\mu \tag{8}$$

4 Executing Casimir's Trick Step by Step

4.1 Step 1: Write Out the Spinor Components Explicitly

Let us write the Dirac indices explicitly. The product $A \times B$ is:

$$A \times B = \Big[\bar{u}_i(4)\ (\Gamma_1)_{ij}\ u_j(1)\Big]\ \Big[\bar{u}_k(1)\ (\Gamma_2)_{k\ell}\ u_\ell(4)\Big] \tag{9}$$

Each factor in square brackets is a 1×1 **number** (a complex scalar), so we can freely rearrange the order of the two brackets:

$$= \bar{u}_i(4)\ (\Gamma_1)_{ij}\ u_j(1)\ \bar{u}_k(1)\ (\Gamma_2)_{k\ell}\ u_\ell(4) \tag{10}$$

Why We Can Rearrange

Each $\bar{u}_i(4)(\Gamma_1)_{ij}u_j(1)$ is a number (the (i,j) element of a matrix contracted with spinor components). Numbers commute, so we can write them in any order. The indices are just labels telling us which spinor component connects to which.

4.2 Step 2: Sum Over Spins of Particle 1

Sum over the two spin states of particle 1 using completeness:

$$\sum_{s_1} u_j^{(s_1)}(1)\ \bar{u}_k^{(s_1)}(1) = (\not{p}_1 + m)_{jk} \xrightarrow{m=0} (\not{p}_1)_{jk} \tag{11}$$

Substituting:

$$\sum_{s_1} A \times B = \bar{u}_i(4)\ (\Gamma_1)_{ij} \underbrace{\Big[\sum_{s_1} u_j(1)\bar{u}_k(1)\Big]}_{(\not{p}_1)_{jk}} (\Gamma_2)_{k\ell}\ u_\ell(4)$$

$$= \bar{u}_i(4)\ (\Gamma_1)_{ij}\ (\not{p}_1)_{jk}\ (\Gamma_2)_{k\ell}\ u_\ell(4)$$

$$= \bar{u}_i(4)\ \Big[\Gamma_1 \not{p}_1 \Gamma_2\Big]_{i\ell}\ u_\ell(4) \tag{12}$$

The completeness relation has "stitched" $u(1)$ and $\bar{u}(1)$ together into $\not{p}_1$, collapsing the two bilinears into **one**.

4.3 Step 3: Sum Over Spins of Particle 4

Now (12) has the form $\bar{u}_i(4)\, Q_{i\ell}\, u_\ell(4)$, where $Q \equiv \Gamma_1 \not{p}_1 \Gamma_2$. Sum over particle 4 spins:

$$\sum_{s_4}\sum_{s_1} A \times B = \sum_{s_4} \bar{u}_i(4)\ Q_{i\ell}\ u_\ell(4)$$

$$= \sum_{s_4} u_\ell(4)\ \bar{u}_i(4)\ Q_{i\ell} \tag{13}$$

The Crucial Rearrangement (Griffiths' "fancy footwork")

In going to (13), we moved $u_\ell(4)$ from the right end to the left. This is allowed because $\bar{u}_i(4) Q_{i\ell} u_\ell(4)$ is a **number** (all indices contracted), and the components u_ℓ and $\bar{u}_i$ are just complex numbers that commute.
But now notice the pattern: $u_\ell(4)\bar{u}_i(4)$ is the (ℓ, i) element of the **outer product** $u(4)\bar{u}(4)$, which is a 4×4 matrix.

Apply completeness for particle 4:

$$\sum_{s_4} u_\ell^{(s_4)}(4)\ \bar{u}_i^{(s_4)}(4) = (\not{p}_4 + m)_{\ell i} \ \xrightarrow{m=0}\ (\not{p}_4)_{\ell i} \tag{14}$$

Substituting:

$$\sum_{s_4}\sum_{s_1} A \times B = (\not{p}_4)_{\ell i}\ Q_{i\ell}$$

$$= (\not{p}_4)_{\ell i}\ (\Gamma_1 \not{p}_1 \Gamma_2)_{i\ell}$$

$$= \sum_{\ell} \Big[\Gamma_1 \not{p}_1 \Gamma_2 \not{p}_4\Big]_{\ell\ell}$$

$$= \mathrm{Tr}\Big[\Gamma_1 \not{p}_1 \Gamma_2 \not{p}_4\Big] \tag{15}$$

In the last step, the sum $\sum_\ell M_{\ell\ell}$ is the **definition of the trace**.

5 The Result

Casimir's Trick Applied to $\mathcal{M}_s\mathcal{M}_t^*$

Substituting $\Gamma_1 = \gamma^\alpha(\not{p}_1 + \not{p}_2)\gamma^\beta$ and $\Gamma_2 = \gamma^\nu(\not{p}_1 - \not{p}_3)\gamma^\mu$:

$$\boxed{\sum_{\text{spins}} A \times B = \text{Tr}\Big[\gamma^\alpha(\not{p}_1 + \not{p}_2)\gamma^\beta \; \not{p}_1 \; \gamma^\nu(\not{p}_1 - \not{p}_3)\gamma^\mu \; \not{p}_4\Big]} \tag{16}$$

Reading left to right inside the trace:

γ^α	from Vertex B of diagram s
$(\not{p}_1 + \not{p}_2)$	propagator of diagram s
γ^β	from Vertex A of diagram s
$\not{p}_1$	completeness of particle 1: $\sum u(1)\bar{u}(1) = \not{p}_1$
γ^ν	from Vertex B of diagram t (conjugated)
$(\not{p}_1 - \not{p}_3)$	propagator of diagram t (conjugated)
γ^μ	from Vertex A of diagram t (conjugated)
$\not{p}_4$	completeness of particle 4: $\sum u(4)\bar{u}(4) = \not{p}_4$

The Pattern: Reading the Trace

The trace reads like a "loop" around the two diagrams stitched together:

$$\Gamma_1 \text{ from } \mathcal{M}_s \;\longrightarrow\; \not{p}_1 \text{ (stitch)} \;\longrightarrow\; \Gamma_2 \text{ from } \mathcal{M}_t^* \;\longrightarrow\; \not{p}_4 \text{ (close the loop)}$$

This is exactly the same pattern as for $|\mathcal{M}_s|^2$ or $|\mathcal{M}_t|^2$, except that Γ_1 comes from one diagram and Γ_2 comes from a *different* diagram. That is what makes this an interference term.

6 Including the Photon Polarization Sums

Now put the trace back into (3) and include the photon sums:

$$\frac{1}{4}\sum_{\text{spins,pol}} \mathcal{M}_s\mathcal{M}_t^* = \frac{e^4}{4st} \underbrace{\sum_\lambda \varepsilon^*_{3\alpha}\varepsilon_{3\nu}}_{-g_{\alpha\nu}} \underbrace{\sum_{\lambda'} \varepsilon_{2\beta}\varepsilon^*_{2\mu}}_{-g_{\beta\mu}}$$
$$\times \; \text{Tr}\Big[\gamma^\alpha(\not{p}_1 + \not{p}_2)\gamma^\beta \; \not{p}_1 \; \gamma^\nu(\not{p}_1 - \not{p}_3)\gamma^\mu \; \not{p}_4\Big] \tag{17}$$

The two factors of $(-g)$ contract α with ν and β with μ:

Final Expression for the Interference

$$\boxed{\frac{1}{4}\sum_{\text{spins,pol}} \mathcal{M}_s\mathcal{M}_t^* = \frac{e^4}{4st}\,\mathrm{Tr}\Big[\gamma^\alpha(\not{p}_1+\not{p}_2)\gamma^\beta\,\not{p}_1\,\gamma_\alpha(\not{p}_1-\not{p}_3)\gamma_\beta\,\not{p}_4\Big]} \tag{18}$$

This is a trace of **8 gamma matrices** (each $\not{p}$ counts as 1). The indices α and β are now contracted (repeated upper/lower), ready for the contraction theorems.

7 Preview: Why This Trace is Zero

As shown in the interference proof document, evaluating this trace proceeds by:

1. Contract α using $\gamma^\alpha(\not{p}_1+\not{p}_2)\gamma^\beta\not{p}_1\gamma_\alpha = -2\not{p}_1\gamma^\beta(\not{p}_1+\not{p}_2)$.
2. Expand $(\not{p}_1+\not{p}_2)(\not{p}_1-\not{p}_3) = -\not{p}_1\not{p}_3+\not{p}_2\not{p}_1-\not{p}_2\not{p}_3$ (using $\not{p}_1^2=0$).
3. Contract β using $\gamma^\beta\not{a}\not{b}\gamma_\beta = 4(a\cdot b)$ on each term:

$$-4(p_1\cdot p_3)+4(p_2\cdot p_1)-4(p_2\cdot p_3) = 2t+2s+2u = 2(s+t+u) \tag{19}$$

4. At $m=0$: $s+t+u=0$, so the entire trace vanishes.

8 Summary: The Anatomy of Casimir's Trick

What Casimir's Trick Does

1. **Start with two spinor bilinears:** $[\bar{u}(4)\Gamma_1 u(1)][\bar{u}(1)\Gamma_2 u(4)]$ — these are numbers.
2. **Write out Dirac indices:** $\bar{u}_i(4)(\Gamma_1)_{ij}u_j(1)\bar{u}_k(1)(\Gamma_2)_{k\ell}u_\ell(4)$.
3. **Sum over particle 1 spins:** $\sum_{s_1} u_j(1)\bar{u}_k(1) = (\not{p}_1)_{jk}$.
   This stitches the two bilinears into one: $\bar{u}_i(4)\big[\Gamma_1\not{p}_1\Gamma_2\big]_{i\ell}u_\ell(4)$.
4. **Rearrange** $u_\ell(4)$ to the front (it is a number).
5. **Sum over particle 4 spins:** $\sum_{s_4} u_\ell(4)\bar{u}_i(4) = (\not{p}_4)_{\ell i}$.
   This closes the loop: $(\not{p}_4)_{\ell i}\big[\Gamma_1\not{p}_1\Gamma_2\big]_{i\ell} = \mathrm{Tr}\big[\Gamma_1\not{p}_1\Gamma_2\not{p}_4\big]$.
6. **No spinors remain.** Everything is now a trace of 4×4 matrices.